

WJRC Computer Services System

Video Presentation Script | Total runtime: approximately 6 to 7 minutes

1. Introduction

0:00 – 0:30

Good day! Today we are presenting our capstone system, the WJRC Computer Services Management System, a web-based platform built to manage the shop's day-to-day operations: repair job orders, point-of-sale transactions, inventory, staff accounts, and sales forecasting, all in one place.

The system was built to replace manual, paper-based tracking with a centralized, role-based platform, so the admin, cashier, and technician all work from the same real-time data, with access limited to what each role actually needs.

On screen: show the login page with the WJRC logo.

2. Sign-Up Process

0:30 – 1:15

WJRC is an internal business system, so there is no open self-registration page. Accounts are created and issued by the Admin through the User Management module, keeping control over who gets access to company data.

On screen: log in as Admin, go to Users, click to add a new user.

The admin fills in the staff member's name, email, and optional username and contact number, then assigns a role — Admin, Technician, or Cashier — and a status, active or inactive.

Notice the required-field validation: name and email are required, and the email and username must be unique across the system. A duplicate is rejected right away. For the password, the system enforces a strong password policy: at least 11 characters, with an uppercase letter, a lowercase letter, a number, and at least two special characters.

On screen: try a weak password to show the validation error, then enter a valid one.

Once saved, the account is created with a hashed password and the correct module permissions automatically applied based on its role, and the action is recorded in the Activity Log for accountability.

3. Login Process

1:15 – 2:00

On screen: log out, go to the login page.

The login form asks for an email and password. Let's first try an incorrect password.

On screen: submit wrong credentials.

The system responds with a generic error, "These credentials do not match our records," without revealing whether the email or the password was wrong. This is a deliberate security choice: it prevents attackers from figuring out which emails are registered in the system. Now with the correct credentials —

On screen: submit correct credentials.

— the user is authenticated and redirected straight to the page their role actually uses. Technicians land on Job Orders, Cashiers land on Inventory and Point-of-Sale, and Admins land on the main Dashboard. Also worth noting: if an account has been deactivated by the admin, login is blocked immediately with a clear message, even if the password is correct.

4. Security Features

2:00 – 3:30

Security was a core design consideration, so let's highlight what's under the hood.

Password protection. Passwords are never stored in plain text. The system uses Bcrypt hashing, and every new or reset password must pass the strong-password rule: minimum 11 characters, mixed case, a number, and two special characters.

Authentication and session handling. After a successful login, the session ID is regenerated to prevent session fixation attacks, and on logout, the session is fully invalidated and the CSRF token is regenerated. Protected pages also send no-cache

headers, so a user cannot hit the browser's back button after logging out and see cached, sensitive pages.

Role-based access control. Every route in the system is protected by role middleware, right down to individual tabs. Only Admins can reach User Management, the Dashboard, or Forecasting. Only Admins and Technicians can reach Job Orders. Admins and Cashiers get full Inventory tools, while Technicians get a read-only Catalog and Archive view plus the ability to submit Parts Out requests. If a user tries to access a page outside their role, they are redirected to their own home page instead of seeing an error page that reveals the app's structure.

On screen: log in as a Technician and try to visit the Users or Dashboard page directly by URL, show the redirect.

Forgot-password flow. Instead of a plain reset link, the system uses a one-time PIN sent by email. The code expires after 2 minutes, allows a maximum of 5 attempts before it is invalidated, and there is a 60-second cooldown before a new code can be requested, protecting against brute-force guessing. The confirmation message is also worded so it never confirms whether an email exists in the system, preventing account enumeration.

Audit trail. Every sensitive action — account creation, role changes, password resets, deactivations, and Parts Out requests, approvals, and rejections — is written to an Activity Log with the staff member, timestamp, and IP address, exportable to CSV for record-keeping.

On screen: show the Activity Log and Password Reset Log screens.

5. Main Dashboard and Modules Overview

3:30 – 4:30

On screen: Dashboard page.

Once logged in as Admin, the main Dashboard shows a real-time snapshot of the business: total revenue, active and pending job orders, total clients, a job-order status breakdown across the full repair pipeline (pending, in progress, for pick-up, completed, cancelled), and a paid-versus-partial payment health bar, plus a demand and revenue trend chart over the last 6 or 12 months.

From the sidebar, the system is organized into five main modules:

1. **Dashboard** — the business overview just shown, admin-only.
2. **Product Transaction / Point-of-Sale** — for processing walk-in sales and billing completed job orders.
3. **Job Orders** — the repair-ticket pipeline, from intake to completion.
4. **User Management** — staff accounts, roles, and the activity and password-reset logs.
5. **Sales Forecasting** — data-driven demand and revenue projections.

Inside Inventory there is also a dedicated Parts Out tab, demonstrated shortly. Each staff member only sees the modules and tabs their role grants. Technicians, for instance, see Job Orders plus a read-only Inventory view and Parts Out, keeping their workspace focused without giving them full stock or pricing control.

6. Key Functionalities and Core Features

4:30 – 6:20

Let's walk through the core features with real examples.

Job Orders is the heart of the repair workflow.

On screen: Job Orders module.

An admin creates a job order by entering the customer's details, their device, and the service needed. The system automatically creates or matches the customer record and assigns a technician. As the technician works on it, they update its progress — pending, in progress, for pick-up, completed — and every status change is logged with a before-and-after record. Once complete, a printable service report can be generated for the customer.

On screen: change a job order's status, then open the service report.

Point-of-Sale handles two kinds of sales in one till: a straight walk-in product sale, or billing an existing job order's labor cost plus any add-on parts. The total is computed server-side as product lines plus labor, so the amount charged cannot be tampered with from the browser.

On screen: process a quick sale.

Parts Out is a newer addition to the system, and it solves a real gap: tracking parts the moment a technician uses them, not just when the customer eventually pays.

On screen: log in as Technician, go to Inventory, Parts Out tab, submit a request against a job order.

A technician picks the job order and the parts used, and submits the request. It goes in as pending approval — stock is not touched yet. Switching to the Admin view —

On screen: switch to Admin, Inventory, Parts Out, approve the pending request.

— the admin can approve or reject it, with a reason required for rejections. On approval, stock is deducted immediately and the part's cost is automatically added to that job order's bill. Once the customer settles at the counter, the admin marks the request as billed. The whole log is filterable and exportable to CSV.

Inventory Management lets admins and cashiers add products, adjust stock and pricing, and archive discontinued items rather than permanently deleting them, preserving sales history. Inventory and sales data can also be exported to or imported from CSV for bulk updates and reporting.

Sales Forecasting is one of the system's standout features. Using historical job-order and sales data, it runs a SARIMA time-series model to forecast demand and revenue up to several months ahead, with a configurable confidence interval. It also surfaces top-performing services and products, day-of-week demand patterns, and holiday impact, giving the owner data to plan staffing and inventory ahead of time rather than reacting after the fact.

On screen: brief look at the Forecasting chart.

User Management and Auditing, touched on earlier, rounds out the system, giving the admin full control over staff accounts and a complete, exportable audit trail of everything that happens in the system.

7. Closing

6:20 – 6:40

To summarize: the WJRC Computer Services Management System brings together secure, role-based account management, a complete repair job-order pipeline with parts tracking, point-of-sale and inventory management, and data-driven sales forecasting, all in a single, auditable platform. Thank you for watching.

On screen: WJRC logo, end card.

Section	Approximate time
Introduction	0:00 – 0:30
Sign-Up Process	0:30 – 1:15
Login Process	1:15 – 2:00
Security Features	2:00 – 3:30
Dashboard and Modules	3:30 – 4:30
Key Functionalities	4:30 – 6:20
Closing	6:20 – 6:40

Total runtime: approximately 6.5 to 7 minutes. Trim the Job Orders or Inventory examples slightly to land closer to 6 minutes — the Parts Out walkthrough is worth keeping full length since it is the newest feature.